to generate traces and send them over to Tempo.

13.4.2 Configuring tracing in Spring Boot with OpenTelemetry

The OpenTelemetry project includes instrumentation that generates traces and spans for the most common Java libraries, including Spring, Tomcat, Netty, Reactor, JDBC, Hibernate, and Logback. The OpenTelemetry Java Agent is a JAR artifact provided by the project that can be attached to any Java application. It injects the necessary bytecode dynamically to capture traces and spans from all those libraries, and it exports them in different formats without you having to change your Java source code.

Java agents are often provided to the application at runtime from the outside. For better dependency management capabilities, in this case, I prefer using Gradle (or Maven) to include the agent JAR file in the final application artifact. Let's see how.

Open your Catalog Service project (catalog-service). Then add a dependency on the OpenTelemetry Java Agent in your build.gradle file. Remember to refresh or reimport the Gradle dependencies after the new addition.

Listing 13.20 Adding dependency for OpenTelemetry Java Agent in Catalog Service

```
ext {
  ...
  set('otelVersion', "1.17.0")                    ❶
}
dependencies {
  ...
  runtimeOnly "io.opentelemetry.javaagent:
  ➥ opentelemetry-javaagent:${otelVersion}"      ❷
}
```